# Identity-Based Cryptography on IoT devices: balancing security and efficiency for modern use cases

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***Abstract*—Identity-Based Cryptography (IBC) provides a revolutionary approach to public-key cryptography, since it enables message encryption without the need of previously distributed keys. Using identities, e.g., email addresses or serial numbers, as public keys, IBC simplifies key management and reduces the reliance on certificate authorities providing a promising lightweight solution for modern Internet of Things (IoT) use cases. Moreover, leveraging elliptic curves and pairings as cryptographic building blocks, Identity-Based Encryption (IBE) offers strong security with small key sizes, which makes it even more attractive for resource-constrained IoT devices. This paper evaluates the resource requirements, i.e., execution time and memory, of IBE protocols by conducting experiments on mid-class IoT devices, i.e., Raspberry Pi3B+ and Pi4B+. We provide insights regarding the potential of IBC for sensitive data confidentiality in modern use cases, such as remote patient monitoring, drone-based data collection and Vehicle-to-Vehicle (V2V) communication.**

***Index Terms*—Identity-Based Cryptography, Elliptic Curves, Internet of Things, Constrained Devices**

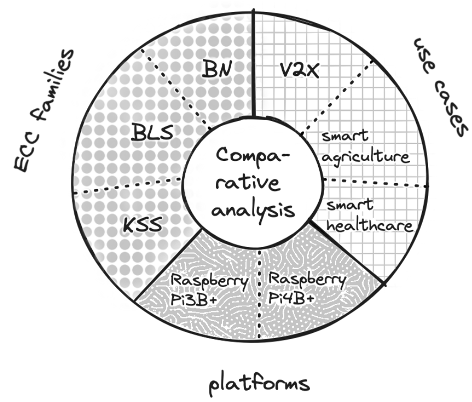



Fig. 1: Identity-based cryptography for modern use cases on IoT constrained devices using different elliptic curves' families.

## I. Introduction

Rapid expansion of the Internet of Things (IoT) in various domains, such as smart healthcare, precise agriculture and Vehicle to Everything (V2X) communication, has led to a significant rise in the number of interconnected devices. In particular, approximately 6 billion IoT devices were deployed in 2018 and, nowadays, it is estimated to reach 18 billion [1]. However, due to the increasing attention they draw, IoT devices are becoming attractive targets for attackers, e.g., the widespread reliance on data collection particularly opens the door for data breaches causing disruptions to critical infrastructure, such as life support systems, agricultural supply chain or traffic control systems, respectively. Therefore, ensuring sensitive data confidentiality is mandatory [2].

To protect IoT platforms and the systems they support is challenging due to their resource-constrained nature. Striking a balance between strong confidentiality and efficiency, in terms of hardware resources, requires lightweight cryptography. Elliptic Curve Cryptography (ECC) and Identity-Based Cryptography (IBC), in particular, is suitable to achieve the aforementioned balance, since it supports a variety of security levels, while maintaining low execution time and memory overhead [3]. Moreover, IBC enables message encryption without the need of previously distributed keys, which is an attractive approach for IoT use-cases, where keys' pre-distribution raises security concerns. It utilizes a trusted third party, i.e., the Private Key Generator (PKG), which authenticates the receiver, using some form of its identity, e.g., email address or devices' IDs, instead of its certificate, generates its private key and provides public system parameters to the sender. IBC has also been used in wireless sensor networks [4].

Regarding platforms utilized in IoT use cases, Raspberry Pi devices are an ideal mid-class solution, since they combine low cost with high processing power and large memory capacity. Their reliability and robustness along with their numerous communication capabilities, I/O options and software support, renders them excellent candidates for smart and autonomous operation in modern healthcare, agriculture and V2X environments where data encryption at rest and in transit is required. For example, sensitive patients' information exchanged among devices require strong confidentiality, i.e., high security level, for data breaches' protection [5], even at the cost of high hardware requirements. In other sectors, such as agriculture, high security level combined with low execution time are essential for timely decision making and data leakage avoidance [6].

On the other hand, time is critical for collision avoidance in Vehicle-to-Vehicle (V2V) communications [7], where low to medium security levels for data encryption is satisfactory.

Fig. 1 illustrates three major aspects regarding cryptography in IoT, i.e., *ECC* offering a lightweight solution for strong confidentiality by leveraging a multitude of configurations provided by ECC families, *IoT platforms* implying constrains in terms of processing and memory resources and, *use cases* along with their associated trade-offs regarding security and performance requirements. Along these lines, we propose and experiment with Identity-Based Encryption (IBE) [8] on low-cost devices to validate it as a feasible solution for future generation services. We argue that the appropriate selection of parameters can bring the balance between security and execution time in a way that suits the demands of the applications in interest. We indicatively refer the domains of smart healthcare, precise agriculture and V2X communication, since a thorough discussion on IoT use cases deserves further study.

Motivated by our previous work [9], the contribution of this paper can be summarized as follows: i) we leverage the advanced features of the Relic-Toolkit library [10] to develop and compare two variations of Boneh and Franklin's IBE protocols, namely *BasicIdent* and *FullIdent*; ii) we study a broader range of elliptic curves derived by the Barreto-Naehrig (BN), Barreto-Lynn-Scott (BLS) and Kachisa-Scott-Schaefer (KSS) families; iii) we conduct experiments on two devices, i.e., Raspberry Pi3B+ and Pi4B+ to evaluate the impact of security level configuration on resource requirements in terms of CPU time and memory; and iv) we bring modern IoT use cases to the foreground highlighting that there is no one-size-fits-all solution and, therefore, the selection of elliptic curve and security level is challenging.

The rest of the paper is organized as follows. Related work is discussed in Section II, while Section III briefly refers indicative use cases highlighting the need of balancing security and efficiency. Section IV provides the fundamentals of pairing operations in IBE schemes, and Section V presents our experimental results. We conclude the paper along with future work insights in Section VI.

## II. Related Work

Future generation networks offer key enabling technologies for ubiquitous IoT deployment [11], where mid-class devices, e.g., Raspberry Pi, are able to handle demanding processes, such as cryptographic operations. Several sectors leverage their capabilities, including healthcare [12], where wearable devices are utilized to monitor patients, agriculture, where weather stations collect data which underpin precision agriculture [6], [13],and V2X communications, where On-Board Units (OBU) are utilized to interconnect vehicles [14]. In all these sectors, lightweight cryptography is crucial for balancing security and efficiency according to requirements [7].

In [15] cryptography in IoT systems is examined across many different low-cost boards for smart environments', power grids' and vehicles applications. However, this work does not delve into cryptographic protocols. Various cryptographic algorithms for enhancing IoT security were investigated in [2], focusing, however, on symmetric cryptography which cannot satisfy the range of requirements supported by Public Key Cryptography (PKC). Both ECC and symmetric cryptography are suggested in [16] as lightweight approach in IoT networks, however, the study does not deepen in the use of different low-cost platforms. The authors in [17] propose Public Key Infrastructure (PKI) in an integrated Cloud Computing and IoT facility to offer confidentiality, but PKI is a costly approach.

The feasibility of elliptic curve cryptography in resource-constrained environments was initially demonstrated in [18], paving the way for further exploration into PKC mechanisms. Both Atribute-Based Encryption (ABE) [19] and IBE [8] leverage elliptic curves, as well as bilinear pairings as the mathematical building block operation. While ABE has received significant attention for its ability to enforce fine-grained access control policies, IBE offers a simpler alternative, particularly suitable for IoT environments where key management of complex attribute sets may be impractical. IBE has been applied in IoT [20], cloud computing, smart homes [21], health care [5] and aerial and ground vehicles [22]. In most cases the evaluation is theoretical or in platforms that do not fit the IoT spectrum. A thorough investigation of the elliptic curves for Pairing-Based Cryptography (PBC) is presented in [3]. While the authors elaborate on a theoretical analysis of all the existing standardized Pairing Friendly (PF) curves, the paper lacks the implementation of relative protocols and the experimentation on low-cost devices.

In our previous work [9] we utilized the elliptic curve BN12 from the BN family to implement the *BasicIdent* and *FullIdent* version, which is chosen-ciphertext secure, of the Boneh and Franklin IBE protocol [8] and, the curve BLS12 from the BLS family for short signature schemes. Our results evaluated execution time, memory and energy consumption on the Raspberry Pi3 platform for $78, 112$ and $160$ security levels. In this work we focus on confidentiality, therefore we compare *BasicIdent* and *FullIdent* protocols, and extend our experiments using a greater range of curves derived by BN, BLS and KSS families, evaluating them on additional security levels (e.g., $78, 112, 128, 160, 192$) and on two IoT devices, i.e., Raspberry Pi3B+ and Raspberry Pi4B+. Beyond these extensions and more importantly, we discuss the connection between the requirements of emerging real-world use cases and the appropriate IBE setup.

## III. Use Cases

Secure and efficient communication among IoT devices is critical to maintain confidentiality of the data collected and exchanged. This is especially important in sectors like healthcare, agriculture and vehicular networks, where data breaches and delays can have severe impact. Ideally, high security level and low execution time are desirable [3], but in practice, as the first increases, so does the latter. Hence, according to the requirements of applications in interest, algorithms' configuration, e.g., defining elliptic curves and security level, and platforms' selection play a crucial role.

Regarding healthcare, high security levels are crucial when sensitive patient information are transmitted between patient's and medical professional's devices [5]. Execution time of the messages' encryption and decryption, in such a scenario, is not a priority. Instead, when an emergency alarm for a patient is dispatched, execution time is of utmost importance. Given that security level requirements and execution time are application dependent, appropriate selection and configuration of the components can facilitate confidential messages' exchange.

Establishing secure communication among IoT devices is imperative in smart agriculture. Proprietary farming practices and strategies should be kept private, hence high security levels must be applied when farmers' operational data are transmitted [13]. On the other hand, accuracy in decision-making is based on field data collection and transmission in real time and with adequate security [13]. In such a case, medium security level, e.g., $128$ could provide the execution time and confidentiality levels being required by the application.

Similarly, V2X communications, demanding high security levels, are those who exchange sensitive data related to private information of vehicle users and their locations, or their security keys [7]. Contrariwise, cooperative collision avoidance and emergency braking prioritise speed to ensure timely responses. In such cases, a low to medium security level is sufficient.

To sum up, by meticulously choosing among elliptic curves, security levels and platforms, it is feasible to strike a balance between security and performance ensuring that vital messages are sent and processed within time and security guarantees for each scenario needed. IBE offers a variety of elliptic curves that can be configured at different security levels, and moreover, exploits the notion of identity to create private keys and release public ones minimizing the cost of key management which is an issue on remote areas with tens to hundreds of IoT devices. Next section presents the fundamentals of IBE and its mathematical foundation, i.e., pairing operations.

## IV. Pairings and Identity-Based Cryptography

Pairings are the mathematical building blocks where the design and implementation of the efficient Boneh and Franklin's IBE protocol are based on [8]. A pairing is a bilinear map $e : G_1 \times G_2 \to G_T$, where $G_1$ and $G_2$ are additive groups, and $G_T$ is a multiplicative group, all of prime order $p$. The properties of pairings include bilinearity (e.g., $e(aP, bQ) = e(P, Q)^{ab}$ for $P \in G_1$, $Q \in G_2$) and non-degeneracy (e.g., $e(P, Q) \neq 1_{G_T}$ for $P \neq O_{G_1}$ and $Q \neq O_{G_2}$). An IBE scheme consists of four algorithms:

- **Setup:** The PKG generates a master key, which is kept secret, and a system's public key ($P_{\text{pub}}$).
- **Extract:** The PKG uses the master key and an entity's identity ($ID$) to generate the entity's private key ($d_{ID}$), which is then distributed to the entity.
- **Encrypt:** Using $P_{\text{pub}}$, the identity ($ID$), and the message ($M$), the encryption algorithm produces a ciphertext ($C$).
- **Decrypt:** The decryption algorithm uses $P_{\text{pub}}$, the entity's private key ($d_{ID}$), and the ciphertext ($C$) to retrieve the original message ($M$) or indicate a failure.

Pairings should be efficient in terms of computational cost and are typically constructed on elliptic curves over finite fields. Asymmetric pairings (where $G_1 \neq G_2$) are generally preferred in practice due to their efficiency. *BasicIdent* and *FullIdent* are two IBE schemes proposed by Boneh and Franklin. *BasicIdent* provides basic security against eavesdropping, but is vulnerable to Adaptive Chosen Ciphertext Attacks (CCA), where an adversary can manipulate ciphertexts to learn information about the plaintext. In *BasicIdent*, a ciphertext $C$ is composed of $(C1, C2)$, where $C2$ is the result of the message $M$ XORed with a hash function output $H$. This makes the ciphertext malleable and susceptible to attacks. *FullIdent* employs the Fujisaki-Okamoto transformation which uses additional hash functions and a final decryption check to make the scheme CCA-secure.

The BN, BLS, and KSS elliptic curves' families are widely used in IBC. BN curves excel in pairing operations and are favored for high performance protocols that do not require top tier security. On the other hand, BLS curves offer a level of security along with efficient computations making them ideal for secure digital signatures and key exchanges. KSS curves are specifically tailored for performance in cryptographic protocols striking a balance between security and computational efficiency. Such curves' families are integral part of many modern cryptographic schemes enabling secure and efficient cryptographic operations [3].

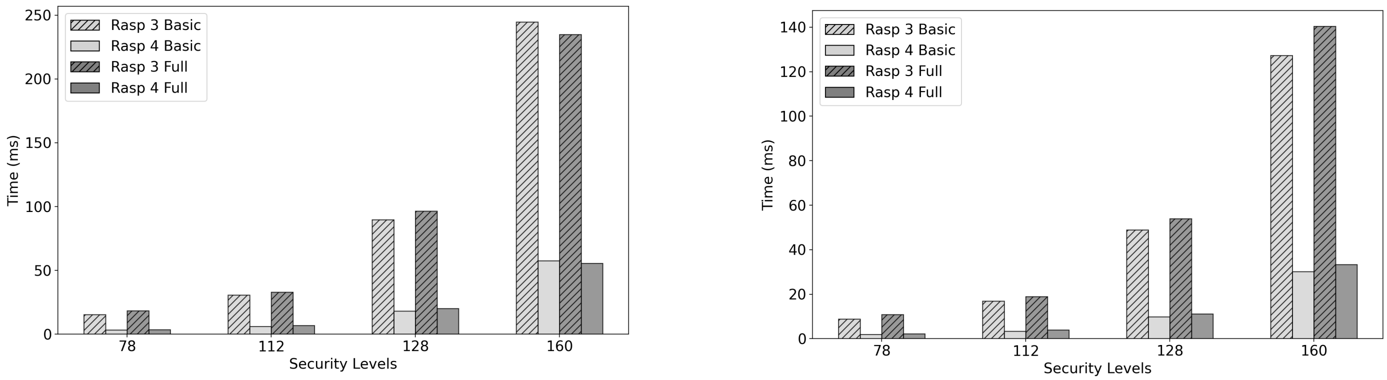
## V. Experimental Results

We conducted our experiments on the Raspberry Pi3B+ and 4B+ platforms running on Raspbian GNU/Linux 11. The first one features a $4$-core ARM Cortex-A53 processor at $1.2$ GHz and 1 GB of RAM, whilst the latter has a $4$-core ARM Cortex-A72 processor at $1.5$ GHz and $4$ GB of RAM. We utilized the Relic-Toolkit library for implementing cryptographic protocols and selected various elliptic curves from the BN, BLS and KSS families. Our source code is available at GitHub[1].

To understand the trade-offs between security and performance in resource-constrained environments, we evaluate and compare *BasicIdent* and *FullIdent* protocols in terms of their time and memory requirements across all combinations of different elliptic curve families, security levels and aforementioned platforms. For statistical reasons the execution time and memory usage were measured over $1000$ runs. The evaluation shows a negligible standard deviation (SD) in execution time, indicating consistent performance; therefore, it is not included in the graphs. On the other hand, for the memory usage, both the mean value and SD were calculated and presented in the corresponding graphs.

Fig. 2 illustrates the execution time required during the encryption and decryption processes on Raspberry Pi3B+ and 4B+ platforms for both *BasicIdent* and *FullIdent* protocols. We select elliptic curves from the BN family, since it supports a range of security levels, i.e., $78, 112, 128$ and $160$. Results, regarding both time and memory, are also derived for $192$

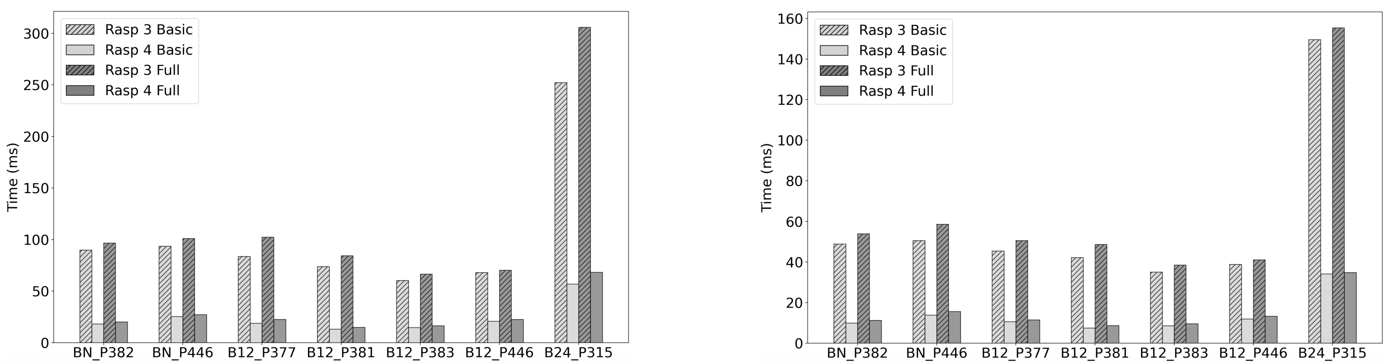
[1]https://github.com/Gavnik/Relic_BonehFranklin




(a) *BasicIdent* and *FullIdent* encryption protocols on Raspberry Pi3+ and Raspberry Pi4+ platforms.

(b) *BasicIdent* and *FullIdent* decryption protocols on Raspberry Pi3B+ and Raspberry Pi4B+ platforms.

Fig. 2: BN curves' family: execution time of pairing-based encryption and decryption for different security levels.




(a) *BasicIdent* and *FullIdent* encryption protocols on Raspberry Pi3B+ and Raspberry Pi4B+ platforms.

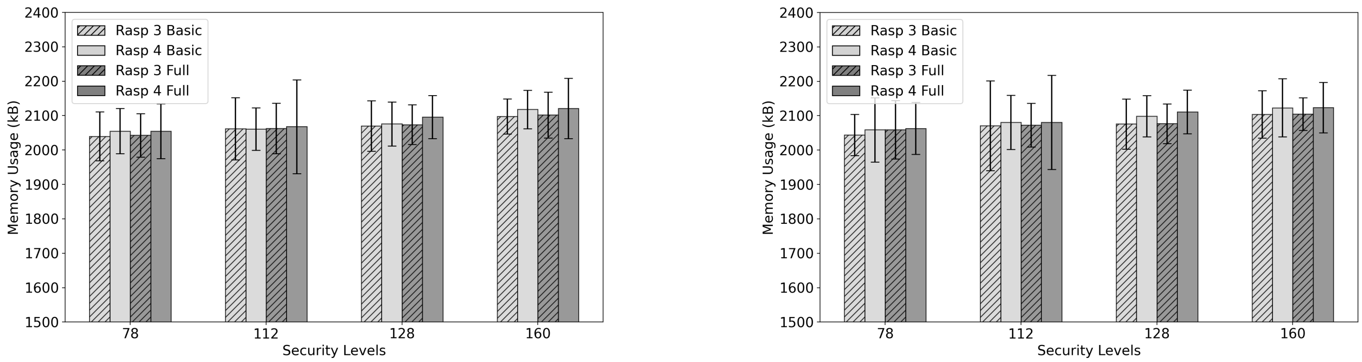
(b) *BasicIdent* and *FullIdent* decryption protocols on Raspberry Pi3B+ and Raspberry Pi4B+ platforms.

Fig. 3: BN and BLS curves' families: execution time of pairing-based encryption and decryption for $128$ security level.

security level which, however, is supported only by BLS and KSS curve families. We omit them due to the lack of space.

In Fig. 2a we observe that as the security level increases, so does the execution time in both platforms, with the $128$ level offering a balance between execution time and security. For example, the execution time on $128$ security level is about $5.5$ and $3$ times higher than $78$ and $112$ security levels, respectively, while tuning at $160$ security level overheads time requirements as much as $13$ and $7$ times compared to $78$ and $112$ security level, correspondingly. Moreover, Raspberry Pi4B+ requires lower execution time compared to Raspberry Pi3B+ across the security levels. This is anticipated since the newer model has faster CPU. A counter-intuitive observation, however, is that while *BasicIdent* on low security levels is faster than *FullIdent* in both platforms, as the security level increases, the gap between them narrows (indicatively from $19.5\%$ at $78$ level down to $7.6\%$ at $128$ level for Raspberry Pi3B+), until the security level $160$, where *FullIdent* becomes faster than *BasicIdent*. This occurs because *BasicIdent* has one mapping that increases in input size with the security level, whereas *FullIdent* has three lower mappings in regards to input size. Consequently, these mappings of *FullIdent* are more efficient than the one of *BasicIdent* at high security levels. For example, the same trend is found for $192$ security level provided by the *K18_P638* curve of the KSS family, where *BasicIdent* requires $401$ and $106$ *ms* on Raspberry Pi3B+ and Pi4B+, accordingly, while *FullIdent* requires $386$ and $103$ *ms*, on the respective platforms (results are not depicted on Fig. 2a since they do not regard the BN family).

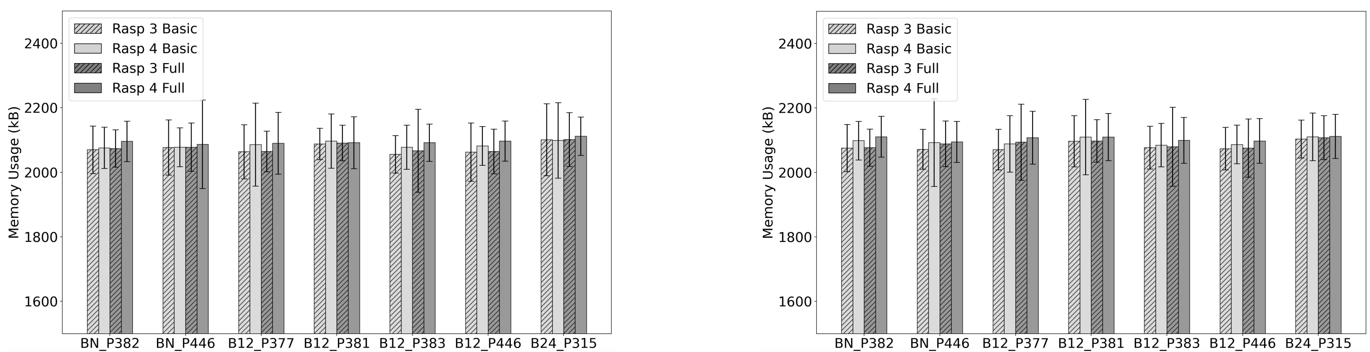
Similarly, Fig. 2b shows that decryption gets slower as security level rises and Raspberry Pi4B+ is faster in decryption than Pi3B+. Regarding the protocols, we observe a corresponding behavior with the increase of the security level, without, however, *BasicIdent* getting slower than *FullIdent*. Similar trends are observed at $192$ security level where the *BasicIdent* on *K18_P638* curve requires $56.7$ *ms* when executed on Raspberry Pi4B+. Comparing Fig. 2a with Fig. 2b we conclude that, in both protocols, encryption is more demanding than decryption, primarily due to the exponentiation overhead during the encryption. Furthermore, *FullIdent* is a better choice than *BasicIdent* as the security level increases, since it produces low time execution overhead (e.g., $21.7\%$ at $78$ level and $10.3\%$ at $160$ level for Raspberry Pi3B+), while at the same time, it offers chosen ciphertext security. In fact, results indicate that




(a) *BasicIdent* and *FullIdent* encryption protocols on Raspberry Pi3B+ and Raspberry Pi4B+ platforms.

(b) *BasicIdent* and *FullIdent* decryption protocols on Raspberry Pi3B+ and Raspberry Pi4B+ platforms.

Fig. 4: BN curves' family: memory usage of pairing-based encryption and decryption for different security levels.




(a) *BasicIdent* and *FullIdent* encryption protocols on Raspberry Pi3B+ and Raspberry Pi4B+ platforms.

(b) *BasicIdent* and *FullIdent* decryption protocols on Raspberry Pi3B+ and Raspberry Pi4B+ platforms.

Fig. 5: BN and BLS curves' families: memory usage of pairing-based encryption and decryption for $128$ security level.

*FullIdent* is the best choice for $160$ security level.

In Fig. 3, we compared different curves at $128$ level because it offers a mid-class solution balancing a high security level without straining hardware resources. This observation is derived from Fig. 2, where between $128$ and $160$ levels the later overheads the execution time heavier compared to low levels, i.e, $78$ and $112$. In addition, it is supported by multiple curves from the BN and BLS families, seven (7) of which are depicted in Fig. 3 and Fig. 5. We note significant disparities in execution times for both encryption and decryption, but in all scenarios, Raspberry Pi4B+ outperforms Pi3B+. Some curves perform exceptionally well (e.g., $B12\_P381$ and $B12\_P383$), while others perform poorly (e.g., $B24\_P315$). Notably, on the Raspberry Pi3B+, $B12\_P383$ gives the best results both on encryption (Fig. 3a) and decryption (Fig. 3b), either for *BasicIdent* or for *FullIdent*, whereas on the Raspberry Pi4B+ the best choice is the curve $B12\_P381$. In both encryption and decryption, $B24\_P315$ gives the slowest execution times that are comparable to that of $160$ security level. In general, at this medium level security, we can conclude that the BLS12 curves perform better than the BN curves, while the BLS24 curves exhibit the worst performance. Results indicate that different elliptic curves and their implementations have a substantial impact on performance, even at the same security level.

Beyond the time-related results, we derive memory requirements of *BasicIdent* and *FullIdent* during encryption and decryption processes on both platforms. Mean value and SD were calculated over $1000$ runs and are depicted in Fig. 4. The results, regarding elliptic curves from the BN family, vary according to the security level selected. Therefore, Fig. 4a shows that both protocols have higher memory demands with the increase of security levels during encryption, while the *FullIdent* protocol is more demanding than the *BasicIdent* protocol at all levels tested. Additionally, the Raspberry Pi4B+ platform consistently require more memory than the Raspberry Pi3B+. The same patterns and conclusions are observed for the decryption process as shown in Fig. 4b.

Elaborating on the $128$-bit security level, Fig. 5 show the memory-related results for both protocols, across seven (7) elliptic curves of BN and BLS families used for encryption and decryption on both platforms. Findings for encryption (Fig. 5a) are identical with those of decryption (Fig. 5b) and can be summarized as follows: the Raspberry Pi3B+ consistently requires less memory than the Raspberry Pi4B+, while between the two protocols under investigation, the *FullIdent* has higher memory demands. Exception to that is observed when we use the B12_P381 curve of the BLS family on Raspberry Pi4B+, where *BasicIdent* requires either more (encryption) memory than or the same (decryption) amount of memory as *FullIdent*.

To conclude, our study reveals that the selected elliptic curve cryptographic protocols can be implemented and executed efficiently on Raspberry Pi platforms. The performance scales according to the hardware capabilities and security requirements. The Raspberry Pi4B+ consistently outperformed the 3B+ in terms of execution time, making it a more suitable choice for applications requiring high security levels. The *FullIdent* protocol, while more secure, incurs minimal additional time overhead compared to the *BasicIdent* protocol and both protocols exhibit moderate increase in memory usage during encryption and decryption as the security level goes higher. Validating feasibility of IBE on Raspberry Pi platforms makes it an attractive candidate for secure and efficient cryptographic operations in mid-class IoT devices. The improvements in hardware capabilities further enhance the protocols' performance, enabling the deployment of secure cryptographic protocols in resource-constrained environments.

## VI. Discussion and Future Work

Evaluating the performance of elliptic curve cryptographic protocols on various hardware platforms provides valuable insights into how these protocols can be effectively implemented on resource-constrained IoT devices. The findings underscore the potential of IBE to offer robust security solutions that are both efficient and scalable.

According to our findings, both Raspberry Pi3B+ and 4B+ are viable options in IoT deployments. In scenarios where execution time is not essential, while security level is vital, either platform could be selected and implement ECC with 160 security level. This way, sensitive patient information, proprietary farming practices and private vehicle information can be adequately safeguarded. When near real time responses are required, Raspberry Pi3B+ marginally qualifies and could be only used with low security level implementation. On the other hand, Raspberry Pi4B+ seems to be an ideal choice when considering efficiency issues, since it requires under 100 *ms* to execute encryption and decryption operations with 160 security level. All security levels tested are feasible, but medium security levels, e.g., 128, manage to better balance execution time and reliable security than the rest. When comes to curve selection, those of the BLS family demonstrate best results in both platforms. Therefore, an IBE setup on Raspberry Pi4B+ using BLS_P381 or BLS_P383 would be recommended on messages serving patient emergency alarm, precision agriculture and cooperative collision avoidance.

Our future work plans include addressing additional levels of confidentiality along with integrity and ability to safeguard future-generation services on IoT constrained devices. For that reason, we aim at extending research on further security levels and implementation of digital signatures with elliptic curves. Furthermore, an extensive analysis on energy consumption of medium and low-cost devices falls into our plans. Finally, optimizing hardware and cryptographic schemes based on budget constraints and the specific requirements of the use cases is to be further examined. This way, we target at an extensive feasibility study, covering all security aspects of future-generation services.

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